

## SEM-4 PROJECT PYTHON+FSD

**Real-Time Problem Solving Project Titles** that combine:

- ✓ React Frontend
  - ✓ Django REST API Backend
  - ✓ Machine Learning / Deep Learning
  - ✓ PostgreSQL/MySQL
  - ✓ Postman Testing
  - ✓ Payment Gateway (Razorpay/UPI/Paytm)
  - ✓ Email Notifications
  - ✓ WhatsApp Integration (Twilio, Meta WhatsApp Cloud API)
  - ✓ SMS Alerts
- 

### 1. Smart Hospital Management and Disease Prediction System

#### Problem

Hospitals struggle with patient management and early disease detection.

#### Features

- Patient registration
- Doctor appointment booking
- Medical history tracking
- Disease prediction using ML

#### Technology

- Frontend: React
  - Backend: Django REST API
  - ML: Random Forest, XGBoost
  - Testing: Postman
- 

### 2. AI-Based Student Performance Prediction System

#### Problem

Educational institutions cannot identify weak students early.

#### Features

- Student records management
- Attendance tracking
- Marks analysis
- Performance prediction

#### ML Algorithm

- Decision Tree
  - Random Forest
  - Neural Networks
- 

### 3. Fake News Detection Platform

#### Problem

Spread of misinformation on social media.

#### Features

- News article submission
- Real/Fake classification
- Dashboard analytics

#### Deep Learning

- LSTM
  - BERT
  - NLP Techniques
-

#### **4. Smart Traffic Congestion Prediction System**

##### **Problem**

Traffic jams in urban areas.

##### **Features**

- Traffic data collection
- Congestion prediction
- Route recommendation

##### **ML Models**

- LSTM
  - CNN-LSTM
- 

#### **5. AI-Based Crop Disease Detection System**

##### **Problem**

Farmers fail to identify crop diseases in time.

##### **Features**

- Upload crop image
- Disease identification
- Treatment suggestions

##### **Deep Learning**

- CNN
  - ResNet50
  - EfficientNet
- 

#### **6. Intelligent Resume Screening System**

##### **Problem**

Manual resume filtering is time-consuming.

##### **Features**

- Resume upload
- Skill extraction
- Candidate ranking

##### **AI Techniques**

- NLP
  - BERT
  - TF-IDF
- 

#### **7. Online Fraud Transaction Detection System**

##### **Problem**

Financial fraud causes huge losses.

##### **Features**

- Transaction monitoring
- Fraud alerts
- Risk scoring

##### **ML Algorithms**

- Isolation Forest
  - Logistic Regression
  - XGBoost
- 

#### **8. Smart E-Commerce Product Recommendation System**

##### **Problem**

Users struggle to find relevant products.

##### **Features**

- Personalized recommendations
- User behavior analysis

- Shopping analytics

#### **ML**

- Collaborative Filtering
  - Deep Neural Networks
- 

### **9. Real-Time Air Quality Monitoring and Prediction System**

#### **Problem**

Pollution monitoring is insufficient.

#### **Features**

- AQI dashboard
- Pollution forecasting
- Health alerts

#### **ML**

- Random Forest
  - LSTM
- 

### **10. AI-Based Mental Health Analysis Platform**

#### **Problem**

Early mental health issues often go unnoticed.

#### **Features**

- Mood tracking
- Sentiment analysis
- Mental health prediction

#### **Deep Learning**

- NLP
  - BERT
  - LSTM
- 

### **11. Smart Attendance System Using Face Recognition**

#### **Problem**

Proxy attendance in institutions.

#### **Features**

- Face capture
- Automatic attendance
- Attendance reports

#### **Deep Learning**

- FaceNet
  - OpenCV
  - CNN
- 

### **12. Real-Time Cyber Attack Detection System**

#### **Problem**

Network intrusions and cyber attacks.

#### **Features**

- Traffic monitoring
- Threat detection
- Alert generation

#### **ML**

- Random Forest
  - Deep Autoencoders
- 

### **13. AI-Powered Job Recommendation Portal**

#### **Problem**

Job seekers struggle to find suitable jobs.

**Features**

- Profile creation
- Skill analysis
- Personalized recommendations

**ML**

- Recommendation Systems
  - NLP
- 

**14. Smart Energy Consumption Prediction System**

**Problem**

High electricity wastage.

**Features**

- Energy monitoring
- Future consumption prediction
- Cost optimization

**ML**

- LSTM
  - Linear Regression
- 

**15. Customer Churn Prediction System**

**Problem**

Companies lose customers without warning.

**Features**

- Customer analytics
- Churn prediction
- Retention recommendations

**ML**

- Random Forest
  - XGBoost
- 

**16. Deep Learning-Based Massive MIMO-OFDM Channel Estimation System**

**Problem**

Traditional channel estimation techniques are less accurate in 6G networks.

**Features**

- OFDM simulation
- Channel estimation
- Performance comparison

**Deep Learning**

- CNN
- DNN
- LSTM

**Suitable for Research Projects**

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**17. AI-Based Disaster Prediction and Alert System**

**Problem**

Delayed disaster warnings.

**Features**

- Weather monitoring
- Flood prediction
- Emergency alerts

**ML**

- LSTM

- Time Series Forecasting
- 

## **18. Smart Waste Management System**

### **Problem**

Inefficient garbage collection.

### **Features**

- Bin monitoring
- Collection route optimization
- Waste prediction

### **ML**

- Predictive Analytics
  - Route Optimization
- 

## **19. AI-Based Loan Approval Prediction System**

### **Problem**

Manual loan approval processes.

### **Features**

- Applicant evaluation
- Risk assessment
- Loan recommendation

### **ML**

- Logistic Regression
  - Random Forest
- 

## **20. Intelligent Road Accident Prediction System**

### **Problem**

Increasing road accidents.

### **Features**

- Accident hotspot analysis
- Risk prediction
- Safety recommendations

### **ML**

- XGBoost
- Random Forest
- Deep Neural Networks

## **1. Smart Government Scholarship Management System**

### **Indian Problem**

Many students miss scholarship opportunities due to lack of awareness and complex processes.

### **Features**

- Student registration
- Scholarship eligibility prediction
- Online application
- Document verification
- Status tracking

### **Integrations**

- Razorpay application fee
- Email notifications
- WhatsApp status updates

### **AI Module**

Scholarship eligibility prediction using ML.

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## **2. Farmer Marketplace and Crop Advisory Platform**

### **Indian Problem**

Farmers often depend on middlemen and lack crop guidance.

### **Features**

- Sell crops directly
- Buy fertilizers/seeds
- Crop disease detection
- Weather alerts

### **Integrations**

- UPI/Razorpay payments
- WhatsApp crop alerts
- Email invoices

### **Deep Learning**

CNN-based crop disease detection.

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## **3. AI-Powered Medical Appointment & Disease Prediction System**

### **Indian Problem**

Rural areas face shortage of doctors and delayed diagnosis.

### **Features**

- Online doctor booking
- Patient records
- Symptom analysis
- Disease prediction

### **Integrations**

- Online consultation payment
- WhatsApp reminders
- Email prescriptions

### **ML Models**

Random Forest, XGBoost.

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## **4. Smart Waste Collection System for Smart Cities**

### **Indian Problem**

Overflowing garbage bins in urban areas.

### **Features**

- Bin monitoring
- Collection scheduling

- Complaint management

#### **Integrations**

- WhatsApp complaint submission
- Email reports

#### **AI Module**

Waste generation prediction.

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### **5. Women Safety and Emergency Response Platform**

#### **Indian Problem**

Safety concerns for women.

#### **Features**

- SOS button
- Live location sharing
- Emergency contacts
- Incident reporting

#### **Integrations**

- WhatsApp emergency alerts
- SMS notifications
- Email reports

#### **AI Module**

Crime hotspot prediction.

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### **6. AI-Based Job Portal for Freshers**

#### **Indian Problem**

Fresh graduates struggle to find suitable jobs.

#### **Features**

- Resume builder
- Skill analysis
- Job recommendations

#### **Integrations**

- Paid premium membership
- WhatsApp interview alerts
- Email notifications

#### **AI Module**

Resume matching and ranking.

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### **7. Smart Traffic Violation and Parking Management System**

#### **Indian Problem**

Illegal parking and traffic congestion.

#### **Features**

- Vehicle detection
- Parking management
- Fine collection

#### **Integrations**

- Razorpay fine payment
- WhatsApp challan alerts
- Email receipts

#### **Deep Learning**

YOLO Object Detection.

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### **8. Local Artisan and Handicraft E-Commerce Platform**

#### **Indian Problem**

Local artisans struggle to reach customers.

## **Features**

- Product selling
- Inventory management
- Order tracking

## **Integrations**

- Razorpay
- WhatsApp order updates
- Email invoices

## **AI Module**

Product recommendation system.

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## **9. Smart Public Complaint Management System**

### **Indian Problem**

Citizens struggle to track government complaints.

### **Features**

- Complaint registration
- Photo uploads
- Tracking dashboard

### **Integrations**

- WhatsApp complaint status
- Email notifications

### **AI Module**

Automatic complaint categorization using NLP.

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## **10. AI-Based Loan Eligibility Prediction System**

### **Indian Problem**

Many people are unaware of loan approval chances.

### **Features**

- Loan application
- Credit analysis
- Approval prediction

### **Integrations**

- Payment for report generation
- Email approval reports
- WhatsApp notifications

### **ML Models**

Random Forest, XGBoost.

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## **11. Smart Blood Bank and Donor Management System**

### **Indian Problem**

Blood shortage during emergencies.

### **Features**

- Donor registration
- Blood stock management
- Emergency requests

### **Integrations**

- WhatsApp donor alerts
- Email campaigns

### **AI Module**

Demand forecasting.

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## **12. AI-Based Placement Management System**

### **Indian Problem**



College placement processes are mostly manual.

**Features**

- Student profile
- Company portal
- Interview scheduling

**Integrations**

- WhatsApp interview alerts
- Email notifications

**AI Module**

Placement probability prediction.

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### **13. Smart Ration Distribution System**

**Indian Problem**

Leakages and inefficiencies in ration distribution.

**Features**

- Beneficiary management
- Distribution tracking
- Inventory monitoring

**Integrations**

- OTP verification
- WhatsApp updates
- Email reports

**AI Module**

Fraud detection.

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### **14. AI-Based Electricity Bill Prediction & Energy Saving Platform**

**Indian Problem**

High electricity consumption.

**Features**

- Usage tracking
- Consumption prediction
- Saving recommendations

**Integrations**

- Online bill payment
- WhatsApp reminders
- Email reports

**ML Models**

LSTM, Linear Regression.

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### **15. Smart Ambulance and Emergency Service System**

**Indian Problem**

Ambulance delays during emergencies.

**Features**

- Ambulance booking
- Live tracking
- Hospital availability

**Integrations**

- Online payment
- WhatsApp updates
- Email confirmations

**AI Module**

Shortest route optimization.

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# Project Title: “AgriSathi”

**AgriSathi: AI-Powered Multilingual AgriTech Platform for Farmers with Crop Disease Detection, Marketplace, Government Scheme Recommendation, and Smart Advisory System**

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## Problem Statement

Indian farmers face several challenges:

- Lack of timely crop disease identification
- Dependence on middlemen for selling produce
- Limited awareness of government schemes
- Language barriers in accessing technology
- Poor market price information
- Lack of expert agricultural guidance

The proposed system provides a single platform where farmers can:

- Detect crop diseases using AI
  - Sell products directly to buyers
  - Receive government scheme recommendations
  - Get weather and market price updates
  - Access information in their preferred language
  - Communicate via WhatsApp notifications
- 

## Technologies Used

### Frontend

- React.js
- Material UI
- Redux Toolkit
- Axios
- React Router

### Backend

- Django
- Django REST Framework
- JWT Authentication

### Database

- PostgreSQL

### Machine Learning / Deep Learning

- TensorFlow
- Keras
- Scikit-Learn
- OpenCV

### APIs

- REST APIs
- Weather API
- WhatsApp Cloud API
- Google Translate API (Optional)

### Payment Gateway

- Razorpay

### Testing

- Postman

### Deployment

**PREPARED BY: SWATI PATEL(SHP)**

- Docker
  - Nginx
  - AWS/Render
- 

## **Main Modules**

### **1. Farmer Registration Module**

#### **Features**

- Mobile Number Registration
- Aadhaar Verification (Demo)
- Profile Management
- Language Selection

#### **Languages**

- English
  - Hindi
  - Gujarati
  - Marathi
  - Tamil
- 

### **2. Crop Disease Detection Module**

#### **Features**

- Upload crop leaf image
- AI-based disease identification
- Treatment suggestions
- Confidence score

#### **Deep Learning Model**

CNN Model

Example Diseases:

| <b>Crop</b> | <b>Disease</b> |
|-------------|----------------|
|-------------|----------------|

|        |              |
|--------|--------------|
| Tomato | Early Blight |
|--------|--------------|

|        |             |
|--------|-------------|
| Tomato | Late Blight |
|--------|-------------|

|        |         |
|--------|---------|
| Potato | Healthy |
|--------|---------|

|      |      |
|------|------|
| Corn | Rust |
|------|------|

|      |       |
|------|-------|
| Rice | Blast |
|------|-------|

#### **Dataset**

PlantVillage Dataset

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### **3. Marketplace Module**

#### **Features**

Farmers can:

- List products
- Set prices
- Upload product photos

Buyers can:

- Search products
- Place orders
- Make payments

#### **Example**

Farmer:

- Wheat
- Rice
- Cotton
- Vegetables

Buyer:

- Direct purchase from farmers
- 

#### **4. Government Scheme Recommendation Module**

##### **Problem**

Farmers are unaware of applicable schemes.

##### **Solution**

Machine Learning recommends schemes based on:

- State
- Land Size
- Crop Type
- Farmer Category
- Income

##### **Example Schemes**

- Pradhan Mantri Kisan Samman Nidhi
- Pradhan Mantri Fasal Bima Yojana
- Kisan Credit Card
- Soil Health Card Scheme

##### **ML Algorithm**

- Random Forest
  - Decision Tree
- 

#### **5. Smart Crop Recommendation System**

##### **Input**

- Soil Type
- Rainfall
- Temperature
- pH Level

##### **Output**

Best Crop Suggestions

Example:

Input:

- Black Soil
- Rainfall 800 mm

Output:

- Cotton
- Soybean
- Groundnut

##### **ML Models**

- Random Forest
  - XGBoost
- 

#### **6. Weather Forecast Module**

##### **Features**

- Current Weather
- Rain Prediction
- Temperature Forecast
- Farming Alerts

##### **Alerts**

- Heavy Rain Warning
  - Heat Wave Warning
  - Irrigation Reminder
-

## 7. Market Price Prediction Module

### Features

- Current Market Rates
- Future Price Prediction

### Example

Today:

- Wheat ₹2500/quintal

Predicted:

- Wheat ₹2800/quintal next month

### Models

- LSTM
  - Time Series Forecasting
- 

## 8. WhatsApp Notification Module

### Notifications

- Order Updates
- Disease Detection Results
- Government Scheme Alerts
- Weather Alerts
- Payment Confirmation

Example:

Hello Ramesh Ji,

Your crop image has been analyzed.

Disease Detected:

Tomato Early Blight

Confidence:

96%

Recommended Treatment:

Use Mancozeb Spray.

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## 9. Email Notification Module

### Features

- Registration Confirmation
  - Order Confirmation
  - Scheme Updates
  - Invoice Generation
- 

## 10. Payment Gateway Module

### Razorpay Integration

### Use Cases

- Marketplace Purchases
  - Premium Advisory Services
  - Soil Testing Requests
- 

### User Roles

#### Farmer

- Upload crops
- Detect diseases
- Sell products

- View schemes

#### **Buyer**

- Purchase products
- Track orders

#### **Agriculture Expert**

- Provide consultation
- Approve recommendations

#### **Admin**

- Manage users
- Manage products
- View analytics

## **1. FarmVerse AI**

### **India's Digital Agriculture Ecosystem**

#### **Problem**

Indian farmers use multiple apps for weather, crop disease, market prices, government schemes, and crop sales.

FarmVerse combines everything into one platform.

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#### **Users**

##### **Farmer**

- Manage crops
- Sell produce
- Apply for schemes
- Get AI recommendations

##### **Buyer**

- Purchase directly from farmers

##### **Agriculture Expert**

- Provide consultations

##### **Admin**

- Manage platform
- 

#### **Modules**

##### **Module 1: Crop Disease Detection**

Upload crop image.

Output:

- Disease name
- Severity percentage
- Treatment suggestions

##### *Deep Learning*

CNN + Transfer Learning

Models:

- MobileNetV2
  - ResNet50
- 

##### **Module 2: Crop Recommendation**

Input:

- Soil Type
- pH
- Rainfall

- Temperature

Output:

- Suitable crops

*ML*

Random Forest

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### **Module 3: Smart Marketplace**

Farmer uploads:

- Wheat
- Cotton
- Rice
- Vegetables

Buyer purchases directly.

*Features*

- Razorpay payment
  - Invoice generation
  - WhatsApp updates
- 

### **Module 4: Government Scheme Recommender**

Input:

- State
- Farmer category
- Land size

Output:

Eligible schemes

*ML*

Decision Tree

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### **Module 5: Yield Prediction**

Predict production before harvesting.

*AI*

LSTM

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### **Module 6: Voice Assistant**

Farmer speaks:

Gujarati:

મારા કપાસમાં રોગ છે?

Hindi:

मेरी फसल में बीमारी है?

System responds in same language.

*AI*

Speech-to-Text

NLP

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### **Research Topics**

1. Crop Disease Detection
  2. Agricultural Recommendation Systems
  3. Market Price Forecasting
- 

### **Startup Potential**

Subscription:

₹99/month

**PREPARED BY: SWATI PATEL(SHP)**

Expert Consultation:  
₹299/session  
Marketplace Commission:  
2%

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## 2. SwasthyaMitra AI

### Rural Healthcare Intelligence Platform

#### Problem

Many villages lack specialists and early diagnosis.

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#### Modules

##### AI Symptom Checker

Input:

- Fever
- Cough
- Headache

Output:

Possible diseases.

*ML*

Random Forest

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##### Medical Appointment System

- Doctor booking
  - Video consultation
- 

##### Medical Report Analysis

Upload:

- Blood Report
- ECG
- X-Ray

*Deep Learning*

CNN

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##### Medicine Reminder

WhatsApp reminders:

Take medicine at 9 AM

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##### Emergency Ambulance Booking

Nearest ambulance detection.

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##### Health Risk Prediction

Predict:

- Diabetes
  - Heart Disease
  - Hypertension
- 

##### Monetization

Consultation Fee:

₹199

Health Packages:

₹499/month

**PREPARED BY: SWATI PATEL(SHP)**



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### 3. RojgarSetu AI

#### Employment and Skill Intelligence Platform

##### Problem

Millions of graduates are unemployed despite vacancies.

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##### Modules

###### Resume Analyzer

Upload resume.

Output:

- ATS Score
  - Missing Skills
  - Improvement Suggestions
- 

###### Skill Gap Analysis

Compare:

- Current skills
  - Industry requirements
- 

###### AI Job Matching

Predict:

- Suitable jobs
  - Expected salary
- 

###### Interview Preparation

Generate:

- Technical Questions
  - HR Questions
- 

###### Certificate Verification

Blockchain-ready module.

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##### AI Models

BERT

XGBoost

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##### Revenue

Premium Membership

₹199/month

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### 4. AgriFinAI

#### AI Credit and Loan Platform for Farmers

##### Problem

Farmers often struggle to obtain loans due to lack of formal credit history.

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##### Modules

###### Farmer Credit Score

Based on:

- Land ownership

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- Yield history
- Sales records

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**Loan Recommendation**

Suggest:

- KCC
- Agriculture Loans

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**Insurance Recommendation**

Recommend suitable crop insurance.

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**Default Risk Prediction**

Predict loan repayment risk.

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**Document Verification**

OCR based.

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**AI**

XGBoost

Random Forest

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**Revenue**

Bank referral commission

Insurance referral commission

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## 5. JanSeva360

**AI-Powered Government Services Platform****Problem**

Citizens often don't know which schemes they qualify for.

**Modules****Scheme Recommendation Engine**

Input:

- Age
- Gender
- Income
- State

Output:

Eligible schemes.

**Document Upload**

Verify:

- Aadhaar
- PAN
- Income Certificate

**AI Chatbot**

Ask:

Which government scheme can I apply for?

**Application Tracking**

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Track status.

### **WhatsApp Updates**

Receive notifications.

### **AI**

NLP + Recommendation Engine